

Anypoint Watchmen

What you need, how to set it up and how to run it, explained step by step for someone who has not done this before.

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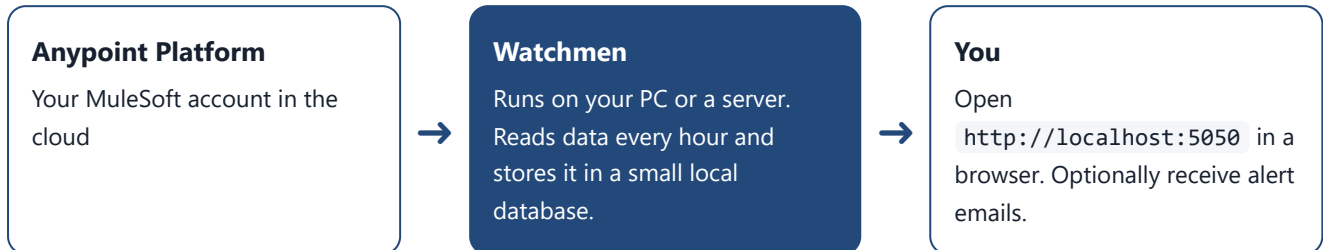
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1. What you are setting up

Watchmen is a small program (a "Mule application") that runs on a computer, wakes up every hour, reads information from your company's MuleSoft Anypoint Platform account and shows it on a web page.



It is safe to try. Watchmen only **reads** from Anypoint. It cannot deploy, change or delete anything in your MuleSoft account.

How long it takes

Task	Who	Time
Install Java and Anypoint Studio (if not already installed)	You / IT	30–45 min
Create the Connected App in Anypoint	Anypoint administrator	15 min
Open the project, add credentials, start it	You	15 min
First-run checks (confirm meter names, contract dates)	You + platform owner	30–60 min

What you will have at the end

- A dashboard in your browser with six tabs: **Overview, Alerts, Audit log, Inventory, Jobs, Settings**.
- Data refreshed automatically every 60 minutes while Watchmen is running.
- Optional: alert emails to a list of people you choose.

2. Shopping list: what you need

Tick these off before you start. Items marked **optional** can be added later.

On the computer

Item	Details / where to get it
☐ A Windows, Mac or Linux computer	8 GB RAM or more, about 3 GB free disk space, and internet access to <code>anypoint.mulesoft.com</code> . A company proxy or firewall must allow it.

☐	Java 17	The language Mule runs on. To check, open a terminal and type <code>java -version</code> ; it should say 17. Install from Oracle or "Eclipse Temurin 17" if missing.
☐	Anypoint Studio 7.x	MuleSoft's free desktop tool for running Mule applications. Download from the MuleSoft website (needs a free MuleSoft account). It includes the Mule 4 runtime.
☐	The Watchmen project folder	The folder named anypoint-watchmen . Copy it anywhere on the computer, for example <code>C:\Projects\anypoint-watchmen</code> .
☐	A web browser	Edge, Chrome or Firefox.
☐	Apache Maven 3.9 optional	Only needed to run the automated tests or build from the command line. Studio already has what it needs.

From your MuleSoft / Anypoint account

Item	Details
☐ An Anypoint administrator	Someone who can open <i>Access Management</i> and create Connected Apps. Step 2 is done by them.
☐ Client ID and Client Secret	The "username and password" of the Connected App created in Step 2. Treat the secret like a password.
☐ Organization ID	A long ID like <code>xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx</code> that identifies your root organisation. Step 2 shows where to find it.
☐ "Usage Viewer" access	Needed to read licence consumption. The administrator grants it to the Connected App.

From the business

Item	Details
☐ Contract figures	The limits, contract start and end month from the latest MuleSoft usage summary (invoice).
☐ Alert recipients	Email addresses of the people who should be told about alerts.
☐ A sending email account optional	For alert emails, for example a Gmail account with an <i>App Password</i> , or your company's SMTP server details.

3. Step by step

1

Install the software

1. Install **Java 17**. Afterwards, open a terminal (Windows: press Start, type *cmd*) and run `java -version`. You should see `17.x`.
2. Download and unzip **Anypoint Studio**. Start it, choose a workspace folder when asked, and sign in with your MuleSoft account if prompted.

2

Create the Anypoint "robot login" (Connected App)

Watchmen logs in to Anypoint by itself, without a person. For that, an **administrator** creates a Connected App:

1. Sign in at `anypoint.mulesoft.com` and open ▶ **Access Management** ▶ **Connected Apps** ▶ **Create app**.
2. Name it **Watchmen Dashboard**. Choose **"App acts on its own behalf (client credentials)"**.
3. Click ▶ **Add Scopes** and add **read-only** scopes for: usage reports (*Usage Viewer*), audit log (*View Audit Logs*), organisation and environments (*View Organization*, *View Environment*), Runtime Manager applications (read/view applications) and API Manager (view APIs). Select the root business group and all environments.
4. Save. Open the app and use ▶ **Copy Id** and ▶ **Copy Secret**. Send them to the person installing Watchmen **securely**, for example through a password manager, never by plain email or chat.
5. Find the **Organization ID**: ▶ **Access Management** ▶ **Business Groups**, then click the top (root) group. The ID is shown on that page and in the browser address bar.

Scope names vary slightly between Anypoint versions. If a later step fails with "403 Forbidden", the Connected App is missing a scope. Ask the administrator to add the one for that area (usage, audit, applications or APIs).

3

Open the project in Anypoint Studio

1. In Studio: ▶ **File** ▶ **Import...** ▶ **Anypoint Studio** ▶ **Anypoint Studio project from File System** ▶ **Next**.
2. For *Project Root*, browse to the `anypoint-watchmen` folder. Click ▶ **Finish**.
3. Wait until the bottom-right progress bar finishes (the first time, Studio downloads components; this can take several minutes).

4

Give it the credentials

The credentials are **not** typed into any project file, so they can never be shared by accident. They are given to Studio when it starts the app:

1. ▶ **Run ▶ Run Configurations....** In the left list, under *Mule Applications*, select **anypoint-watchmen**. If it is not there, double-click *Mule Applications* to create it and tick the project.
2. Open the ▶ **Arguments** tab and add these three lines to the arguments box, replacing the parts in angle brackets with the values from Step 2:

```
-M-Danypoint.client.id=<your client id>  
-M-Danypoint.client.secret=<your client secret>  
-M-Danypoint.orgId=<your organization id>
```

3. Click ▶ **Apply**.

5

Start it and open the dashboard

1. In the same window click ▶ **Run**. Alternatively, right-click the project, then ▶ **Run As ▶ Mule Application**.
2. Watch the *Console* at the bottom. When you see **DEPLOYED** next to *anypoint-watchmen*, it is running.
3. Open a browser at `http://localhost:5050` .

The first automatic data collection starts about **1 minute** after start-up and then repeats every 60 minutes. You do not have to wait. Open the **Jobs** tab and click **Run collect_all**, then refresh after about 30 seconds.

Watchmen only collects while it is running. If the computer sleeps or Studio is closed, collection pauses. On the next run, the audit log continues from where it stopped, looking back up to 24 hours by default.

6

First-run checks (do these once)

a) Check the job worked. In the **Jobs** tab, the latest *collect_all* row should say **success**. If it says **partial** or **failed**, the *Result* column says which part failed. See Troubleshooting.

b) Confirm the usage meter names. Anypoint names each usage measurement (a "meter"). Only some of the names in Watchmen's settings file are confirmed. To see the real names for your account, open this address in the browser while Watchmen is running:

```
http://localhost:5050/api/usage/meters
```

Then, in Studio, open `src/main/resources/config/entitlements.json`. For each line whose `"verified"` is `false`, set `meter` and `measurement` to the matching names from that list and change `verified` to `true`. Ask your MuleSoft account team if a name is unclear.

c) Enter the contract details. In the same file, set the `contract` section (`id`, `start`, `end` as `YYYY-MM`) and check every `limit` against the invoice.

d) Restart the app (red stop button, then Run again) so the changes take effect, and run *collect_all* once more.

4. Optional: email alerts

- In `src/main/resources/config/app.yaml`, under `email:`, set `enabled: "true"`, `from:` to the sending address and `smtp.user:` to the account username. Gmail defaults are already filled in (`smtp.gmail.com`, port `465`).
- Add the email password as one more run argument (Step 4). For Gmail this must be an **App Password** created in the Google account's security settings:

```
-M-Demail.smtp.password=<app password>
```

- Restart, then open the dashboard **Settings** tab, enter the recipients (one per line) and click **Save settings**.

Emails list every new or escalated alert since the previous email. If sending fails, Watchmen tries again on the next run, so nothing is lost.

Settings you can change, and where

Setting	Default	Where to change
Alert recipients	none	Dashboard → Settings
Audit actions ignored	login, logout, read, view, list, search, get, permissions change	Dashboard → Settings

Auto-close audit alerts after	7 days	Dashboard → Settings
Warning / Critical threshold	80% / 95%	<code>app.yaml</code> → <code>alerts</code>
Collection frequency	60 minutes	<code>app.yaml</code> → <code>scheduler</code>
Dashboard port	5050	<code>app.yaml</code> → <code>http.port</code>
Which changes are Critical / Warning	predefined rules	<code>config/audit_rules.json</code>
Contract limits	from invoice	<code>config/entitlements.json</code>

5. Everyday use

Tab	What it shows / what to do
Overview	Alert counts and the <i>Contract entitlements</i> table: used, limit, % and status (OK / WARNING / CRITICAL). Below it, monitoring-only metrics that never alert.
Alerts	All alerts. Click Acknowledge when someone is looking at it and Resolve when it is dealt with. Licence alerts also close themselves when usage drops.
Audit log	Significant changes made in Anypoint (who, what, when), filterable by Critical / Warning / Info. Below it, a history of each hourly audit check.
Inventory	Applications, APIs, environments and business groups found in the latest collection.
Jobs	History of every collection run, plus buttons to run one now.
Settings	Recipients and audit options (editable), plus read-only values from <code>app.yaml</code> .

6. Troubleshooting

What you see	What it means / what to do
App will not start: <i>"Couldn't find configuration property value for key anypoint.client.id"</i>	The run arguments from Step 4 are missing or mistyped. Each line must start with <code>-M-D</code> .
<i>"Address already in use"</i> / port 5050	Another program uses port 5050. Change <code>http.port</code> in <code>app.yaml</code> (for example to 5051) and use that in the browser address.
Jobs tab: failed with 401 or <i>"no access_token"</i>	Client ID or Secret is wrong, or the Connected App was deleted or its secret regenerated. Re-copy them (Step 2).
Jobs tab: 403 <i>Forbidden</i> on usage, audit, apps or APIs	The Connected App lacks a scope for that area. The administrator adds it. Other parts keep working meanwhile.

Some meters listed as failed, or showing 0	The meter name in <code>entitlements.json</code> is not right for your account. Do first-run check 6b. Remember Anypoint usage data is about 3 days behind.
Audit log tab is empty	Normal if nobody changed anything significant recently: routine events are deliberately ignored. Check the <i>Recent audit polls</i> table shows success .
No emails arrive	Check <code>email.enabled</code> is <code>"true"</code> , recipients are saved, and the password argument is set. The Studio Console shows <i>"Alert email failed: ..."</i> with the reason. Check spam folders too.
Dashboard shows "API unreachable"	Watchmen is not running. Start it again in Studio (Step 5).

Studio's Console shows everything Watchmen does. Lines starting with `watchmen.` are the most useful. When running outside Studio, the same lines go to `logs/anypoint-watchmen.log`.

7. Running on a server or in the cloud

Running in Studio on a laptop is ideal for trying Watchmen out. For continuous monitoring it should run somewhere that is always on. Your IT or MuleSoft team can do this:

Option A: CloudHub 2.0 (recommended)

- Deploy the app from Studio or Runtime Manager to a small (0.1 vCore) replica.
- Enter the credentials in Runtime Manager → *Properties* and mark them **secure**.
- Use a **PostgreSQL** database instead of the built-in file database, which does not survive restarts on CloudHub. Set `db.url`, `db.driver` (`org.postgresql.Driver`), `db.user`, `db.password`.
- Note: one Watchmen replica consumes a small amount of your own vCore entitlement.

Option B: an on-premises server

- Install a Mule 4.8 runtime (or Studio) on an always-on server with Java 17.
- Build the package with `mvn clean package` and copy the `.jar` from `target/` into the runtime's `apps` folder.
- Pass the same `-M-D...` values when starting the runtime.
- Keep the dashboard on the internal network: it has no login yet.

8. Where things live (for IT)

Path in the project	Purpose
<code>src/main/mule/*.xml</code>	The application logic: login (auth), usage, audit, alerts, inventory, jobs, settings, web API.
<code>src/main/resources/config/app.yaml</code>	Port, database, schedule, thresholds, email server.
<code>src/main/resources/config/entitlements.json</code>	Contract limits and usage meter names.

<code>src/main/resources/config/audit_rules.json</code>	Which audit changes are Critical or Warning.
<code>src/main/resources/web/index.html</code>	The dashboard page.
<code>src/test/munit/</code>	Automated tests. Run with <code>mvn clean test</code> . They use fake data and need no Anypoint access.
<code>~/watchmen-data/</code> (your home folder)	The local database file created when running in Studio. Delete it to start fresh.

Security notes

- Never put the Client Secret or email password in any file in the project. Use run arguments or secure properties only.
- If a secret is ever shared by mistake (email, chat, a document), ask the administrator to **regenerate** it in the Connected App.
- Give the Connected App read-only scopes only. Watchmen needs nothing else.

9. Glossary

Term	Plain-language meaning
Anypoint Platform	MuleSoft's website and cloud service where integrations and APIs are built, deployed and managed.
Anypoint Studio	The free desktop program used to open and run Mule projects such as Watchmen.
Mule application	A program that runs on the Mule runtime. Watchmen is one.
Connected App	A "robot" login in Anypoint that a program uses instead of a person. It has a Client ID and a Client Secret.
Client ID / Client Secret	The username and password of the Connected App.
Organization ID	The unique ID of your company's top-level Anypoint account.
Business group	A sub-division of the Anypoint account, such as a team or department.
Environment	A separate space such as Sandbox or Production inside a business group.
Entitlement	Something the contract allows a fixed amount of, such as 280 Mule Flows.
Meter	Anypoint's name for one thing it measures, such as the number of flows or messages.
High watermark	Billing on the <i>peak</i> amount reached, for example the most flows running at once.
Drawdown	A pool used up over the contract period, for example 47 million messages in total.
Audit log	Anypoint's record of who changed what, and when.

Job	One collection run. <i>collect_all</i> runs every hour; others can be started from the Jobs tab.
localhost:5050	"This computer, door number 5050": the address of the dashboard while Watchmen runs on your machine.
Run argument (-M-D...)	A value handed to the app when it starts, used here to pass secrets without saving them in files.

Quick-start recap: install Java 17 and Studio → admin creates Connected App → import project → add the three `-M-D` arguments → Run → open `http://localhost:5050` → Jobs → *Run collect_all* → confirm meter names and contract dates.